NEURALPATH CHEATSHEET

Module 1: Foundations

Build intuition for models, vectorized Python, NumPy, Pandas, linear algebra, multivariable cal

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Core Formulas, Mental Models & Engineering Rules

- Model: y_hat = f(x; theta), trained to minimize empirical loss L(theta).
- Vectorization: prefer NumPy matrix operations (X @ W + b) over Python loops.
- Linear Algebra: dot product u.v = ||u|| ||v|| cos(theta); SVD factors M = U Sigma V^T.
- Calculus: gradient points uphill; chain rule propagates partial derivatives backward.
- Information Theory: Cross-Entropy H(p, q) = -sum p(x) log q(x) = H(p) + KL(p || q).

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Estimated study time: 4 hours · Badge: Mathematical Thinker
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